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Course: Calculus Beta

Assignment: MyLab Math - Uge 9

The length of time (in years) until a particular radioactive particle decays is a random variable t with probability density function defined by $f(t) = 11e^{-11t}$ for t in $[0, \infty)$. Find the mean of the distribution, the standard deviation of the distribution, and the probability that the random variable is between the mean and 1 standard deviation above the mean.

The probability density function given is an exponential distribution. The mean of an exponential distribution ae^{-at} is given by $\mu = \frac{1}{a}$.

For the given function, the value of a is 11.

Substituting $a = 11$ into the formula, we see that the mean of the distribution is $\frac{1}{11}$ yrs.

Similarly, the standard deviation of an exponential distribution is given by $\sigma = \frac{1}{a}$. Substituting $a = 11$, we see that the standard deviation is also $\frac{1}{11}$ yrs.

To determine the probability that the random variable is between the mean and 1 standard deviation above the mean, we need to integrate the probability density function from $\mu = \frac{1}{11}$ to $\mu + \sigma = \frac{1}{11} + \frac{1}{11} = \frac{2}{11}$.

Now integrate $f(x)$ from $\frac{1}{11}$ to $\frac{2}{11}$. First, factor 11 out of the integral.

$$\int_{1/11}^{2/11} 11e^{-11t} dt = 11 \left(-\frac{1}{11} e^{-11t} \right) \Bigg|_{1/11}^{2/11}$$

Factoring $-\frac{1}{11}$ out of the expression and substituting the limits of integration gives $-(e^{-2} - e^{-1}) \approx 0.23$. Therefore, the probability that the random variable is between the mean and 1 standard deviation above the mean is 0.23.